

DECENTRALIZED ZERO TRUST NETWORK ACCESS WITH AI SECURITY

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DECLARATION

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ABSTRACT

Decentralized Zero Trust Network Access with AI Security

Contemporary educational IoT network architectures predominantly rely on centralized authentication and access control frameworks that expose critical single points of failure and introduce significant latency overhead. This research presents a Decentralized Zero Trust Network Access (DZTNA) system that addresses these fundamental vulnerabilities through the integration of customized Hyperledger Fabric blockchain for immutable device identity management, Software-Defined Networking (SDN) enforcement via the Ryu controller, and TensorFlow Lite-powered AI anomaly detection for real-time threat identification and mitigation.

The proposed system implements a Zero Trust security model in which every network device is continuously verified before being granted access, with no entity receiving implicit trust regardless of its network position. A Python/Flask blockchain ledger employing SHA-256 cryptographic block linking maintains a tamper-proof single source of truth for device authorization status. The Ryu-Blockchain middleware intercepts OpenFlow Packet-In events, extracts MAC addresses, and performs real-time REST API validation against the distributed ledger, enforcing immediate packet denial for unauthorized devices. AI-based anomaly detection running on edge hardware monitors network flow patterns and generates alerts for DDoS attempts, unauthorized access patterns, and behavioral deviations.

Experimental evaluation demonstrates that the system eliminates the 50% latency increase associated with cloud-dependent authentication while maintaining offline resilience through local blockchain replication. The architecture achieves sub-100ms authentication decisions for registered devices and provides tamper-proof audit trails for all access events. Comparison against traditional centralized authentication systems shows a 78% reduction in authentication-related single-point-of-failure incidents and near-zero false negative rates for unauthorized device detection in controlled test environments. These findings demonstrate that decentralized Zero Trust principles are technically feasible in resource-constrained educational settings, offering a robust, privacy-preserving security framework for IoT-enabled smart classrooms.

Keywords: Zero Trust Architecture, Hyperledger Fabric, Software-Defined Networking, AI anomaly detection, IoT security, blockchain authentication

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BLE	Bluetooth Low Energy
CSNE	Computer Systems and Network Engineering
DDoS	Distributed Denial of Service
DZTNA	Decentralized Zero Trust Network Access
ECC	Elliptic Curve Cryptography
FL	Federated Learning
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
MAC	Media Access Control
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NIST	National Institute of Standards and Technology
OpenFlow	Open Flow Protocol
QoS	Quality of Service
REST	Representational State Transfer
SDN	Software-Defined Networking
SHA	Secure Hash Algorithm
SLIIT	Sri Lanka Institute of Information Technology
TLS	Transport Layer Security
ZTNA	Zero Trust Network Access

CHAPTER 1: INTRODUCTION

1.1 Background Literature

1.1.1 The Rise of IoT-Enabled Smart Classrooms

The integration of Internet of Things (IoT) devices into educational environments has accelerated dramatically over the past decade, driven by the proliferation of affordable sensor hardware, ubiquitous wireless connectivity, and growing institutional demand for data-driven classroom management. Smart classroom systems now routinely incorporate IP cameras for behavioral analysis, environmental sensors for comfort monitoring, NFC readers for attendance tracking, and edge computing gateways for on-premises data processing. Surveys of IoT deployment in higher education institutions across developing regions indicate that over 65% of newly constructed campus facilities incorporate IoT-enabled monitoring infrastructure as of 2024, with projections suggesting near-universal adoption by 2030 [1].

However, the rapid expansion of IoT device fleets in educational settings has outpaced the development of commensurate security frameworks. A classroom deploying twenty IoT sensors, four IP cameras, and a Raspberry Pi edge gateway represents an attack surface comprising potentially hundreds of network endpoints, each presenting its own authentication and authorization requirements. Traditional network security approaches, designed for environments with a small number of well-managed endpoints, are poorly adapted to the scale, heterogeneity, and resource constraints characteristic of educational IoT deployments [2].

1.1.2 Centralized Authentication: Limitations and Vulnerabilities

The dominant paradigm for IoT network security in educational settings relies on centralized authentication servers that validate device identities and issue access tokens. Under this model, a cloud-hosted Identity and Access Management (IAM) service maintains a registry of authorized devices and processes authentication requests from all network endpoints. While conceptually straightforward, this architecture introduces several critical vulnerabilities that are particularly acute in educational IoT environments.

The most significant structural weakness of centralized authentication is the single point of failure it creates. When the central authentication server becomes unavailable — whether due to network partition, hardware failure, or deliberate denial-of-service attack — all devices dependent on it lose the ability to authenticate, rendering the entire smart classroom network inoperative. Empirical studies of educational institution network outages attribute 43% of total downtime to authentication service failures, underscoring the operational risk of centralized dependencies [3]. Furthermore, cloud-dependent authentication introduces latency that is unacceptable for time-sensitive IoT operations. The round-trip time for a device to authenticate against a remote cloud server introduces 50–200 ms of additional latency per access decision, which cumulates to significant delays in systems requiring continuous re-authentication of active sessions [4].

The security posture of centralized authentication systems is further compromised by their implicit trust model. Once a device has authenticated and received an access token, traditional systems typically grant it unrestricted access to all permitted resources until the token expires. This perimeter-based trust model fails to account for the possibility that an authenticated device may subsequently be compromised, its credentials stolen, or its behavior hijacked by malware operating within the trusted perimeter [5].

1.1.3 Zero Trust Security Principles

The Zero Trust security model, first articulated by Forrester Research in 2010 and subsequently formalized by NIST in Special Publication 800-207 (2020), fundamentally rejects the perimeter-based trust model in favor of continuous verification. The core principle of Zero Trust — 'never trust, always verify' — mandates that no network entity receives implicit trust based on its network location, prior authentication, or device type. Every access request must be explicitly authenticated, authorized, and continuously validated throughout its session [6].

NIST SP 800-207 identifies seven foundational tenets of Zero Trust Architecture (ZTA): all data sources and computing services are treated as resources; all communication is secured regardless of network location; access to individual resources is granted on a per-session basis; access is determined by dynamic policy including observable state of client identity and behavior; the enterprise monitors and measures the integrity and security posture of all owned and associated assets; all resource authentication and authorization are dynamic and strictly enforced before access is allowed; and the enterprise collects as much information as possible about the current state of assets and uses it to improve security posture [6].

Implementation of Zero Trust in IoT environments presents particular challenges not fully addressed by NIST SP 800-207, which was designed primarily for enterprise IT contexts. IoT devices typically lack the computational resources to run conventional TLS certificate management stacks, the persistent storage required for local certificate caches, or the battery capacity to sustain continuous cryptographic handshake operations. These constraints necessitate lightweight Zero Trust implementations adapted specifically for resource-constrained IoT hardware [7].

1.1.4 Blockchain for Decentralized Identity Management

Blockchain technology offers a compelling solution to the single-point-of-failure vulnerability inherent in centralized authentication by distributing the device identity registry across multiple nodes in a tamper-proof ledger. Distributed ledger technology eliminates the central server dependency by replicating the complete device authorization state across all participating nodes, ensuring that the failure of any individual node does not interrupt authentication services [8].

Hyperledger Fabric, an enterprise-grade permissioned blockchain platform developed under the Linux Foundation's Hyperledger project, has emerged as the dominant framework for enterprise IoT identity management. Unlike public blockchains such as Bitcoin or Ethereum, Hyperledger Fabric is designed specifically for private, permissioned environments where all participants are known and trusted, transaction throughput requirements are high, and regulatory compliance demands auditability. Hyperledger Fabric's modular architecture supports pluggable consensus mechanisms, enabling deployment configurations optimized for the latency and throughput requirements of IoT authentication workloads [9].

Previous research on blockchain-based IoT authentication has demonstrated that distributed ledger approaches can achieve authentication latency comparable to centralized systems while eliminating single-point-of-failure vulnerabilities. Chen et al. (2022) demonstrated blockchain-based device authentication for industrial IoT achieving 95ms median latency with zero authentication failures during simulated server outages [10]. However, full Hyperledger Fabric deployments require significant computational

resources, and simplified custom blockchain implementations optimized for edge device constraints represent an important research direction for educational IoT security [11].

1.1.5 Software-Defined Networking for Policy Enforcement

Software-Defined Networking (SDN) decouples the network control plane from the data plane, enabling centralized, programmable control of network forwarding behavior through standardized protocols such as OpenFlow. In the context of IoT security, SDN enables fine-grained, dynamic enforcement of access control policies at the network level, complementing application-layer authentication mechanisms with infrastructure-level controls [12].

The Ryu SDN controller, an open-source Python-based SDN framework, provides a flexible platform for implementing custom network security applications. Ryu's event-driven architecture allows applications to intercept OpenFlow Packet-In events, inspect packet headers, and issue flow modification commands to OpenFlow-enabled switches in real time. This capability enables the implementation of MAC-address-based access control that can enforce blockchain-validated authorization decisions at the hardware level, denying network access to unauthorized devices at the switch level before their traffic reaches any application layer [13].

The combination of blockchain-based identity management with SDN-enforced access control creates a defense-in-depth security architecture that addresses vulnerabilities at multiple layers. The blockchain layer provides tamper-proof device identity records; the SDN layer enforces access decisions at network speed; and the AI layer provides behavioral anomaly detection to identify compromised but ostensibly authorized devices [14].

1.1.6 AI-Based Anomaly Detection in IoT Networks

Machine learning-based anomaly detection has emerged as an essential complement to authentication-based security controls, addressing the limitation that authorized devices can exhibit malicious behavior after successfully authenticating. Neural network-based classifiers trained on normal network flow patterns can identify statistical deviations indicative of DDoS participation, data exfiltration, command-and-control communication, and lateral movement attacks with high accuracy and low false positive rates [15].

TensorFlow Lite, Google's inference-optimized deployment framework for neural network models on edge and mobile hardware, enables deployment of trained anomaly detection models on resource-constrained edge devices such as Raspberry Pi without requiring cloud inference infrastructure. Quantization techniques reduce model size by 4-8x with minimal accuracy degradation, enabling models with 90%+ test accuracy to run at 15-30ms inference latency on Cortex-A53 processors [16].

The combination of supervised classification (trained on labeled attack traffic datasets) with unsupervised anomaly detection (statistical deviation from baseline network behavior) provides robust coverage against both known attack patterns and zero-day threats. Federated learning approaches, in which local models are trained on device-specific traffic patterns and periodically aggregated without sharing raw traffic data, offer a privacy-preserving mechanism for continuous model improvement across multiple classroom deployments [17].

1.2 Research Gap

The literature review identifies several critical gaps in existing approaches to IoT network security in educational environments that motivate this research.

System	ZT Model	Blockchain Auth	SDN Enforcement	AI Anomaly	Offline Resilience	Edge Deployment
Centralized IAM (2020-2024)	No	No	No	No	No	No
NIST ZTA SP800-207	Yes	No	Partial	No	No	No
Chen et al. (2022) - BC-IoT	Partial	Yes	No	No	Yes	No
Mininet SDN Controller	No	No	Yes	No	N/A	Yes
Azure IoT Hub + Defender	Partial	No	No	Yes	No	Partial
This Research (DZTNA)	Full	Yes	Yes	Yes	Yes	Yes

Table 1.1: Comparison of Existing IoT Network Security Systems

The gap analysis reveals four primary deficiencies in existing systems. First, no existing solution designed for resource-constrained educational IoT deployments combines all three security layers: blockchain-based decentralized identity, SDN-enforced access control, and AI-driven behavioral anomaly detection. Commercial solutions such as Azure IoT Hub Defender provide AI-based anomaly detection and partial Zero Trust enforcement but require cloud connectivity and impose per-device subscription costs that are prohibitive for developing-region educational institutions operating on constrained budgets.

Second, existing blockchain-based IoT authentication frameworks have been designed for industrial or enterprise contexts with adequate computational resources, reliable network connectivity, and dedicated IT staff. The specific constraints of educational IoT — heterogeneous student-owned and institution-owned devices, intermittent internet connectivity, absence of dedicated network security personnel, and sensitivity of student behavioral data — have not been adequately addressed.

Third, the integration of Zero Trust principles with SDN policy enforcement at the OpenFlow level, using blockchain-validated device identity as the trust anchor, represents a novel architectural combination not present in existing literature. Prior work has examined blockchain-SDN integration for enterprise traffic engineering but not for Zero Trust access control in IoT environments.

Fourth, AI anomaly detection for IoT classroom networks has not been evaluated in the context of the specific traffic patterns generated by multi-modal student monitoring systems, which combine behavioral analysis data streams, environmental sensor telemetry, attendance records, and network management traffic in a single shared network.

1.3 Research Problem

Centralized authentication architectures for IoT-enabled smart classroom networks are fundamentally vulnerable to single-point-of-failure attacks, introduce unacceptable

authentication latency for real-time IoT operations, and fail to detect post-authentication security breaches resulting from device compromise. Approximately 80% of deployed educational IoT systems depend on cloud-hosted authentication services, creating latency increases of up to 50% relative to local processing and introducing privacy risks for sensitive student behavioral data transmitted to external cloud providers.

Traditional perimeter-based security models are inadequate for the dynamic, heterogeneous device environments characteristic of smart classrooms, where the attack surface spans student-owned devices, institutional IoT sensors, edge computing gateways, and network infrastructure components, each potentially controlled by different stakeholders with varying security postures. The absence of continuous device verification mechanisms means that a single compromised device within the trusted perimeter can execute lateral movement attacks, data exfiltration, or DDoS participation without triggering security alerts.

Existing solutions do not simultaneously address the three critical requirements for secure, resilient, cost-effective smart classroom IoT networks in resource-constrained educational environments: (1) tamper-proof, decentralized device identity management that maintains operational continuity during network partitions; (2) zero-trust access control enforced at the network infrastructure layer; and (3) AI-driven behavioral anomaly detection capable of identifying compromised devices that have successfully authenticated.

1.4 Research Objectives

Main Objective

To design, implement, and evaluate a Decentralized Zero Trust Network Access system that provides resilient, tamper-proof authentication and AI-enhanced behavioral security for IoT-enabled smart classroom networks, operating effectively in resource-constrained educational environments without dependency on cloud authentication infrastructure.

Sub-Objectives

- To design and implement a customized blockchain ledger using SHA-256 cryptographic block linking that maintains a tamper-proof single source of truth for device authorization status, supporting both online synchronization and offline-resilient local operation.
- To develop a Ryu-Blockchain middleware layer that intercepts OpenFlow Packet-In events, performs real-time blockchain authorization queries via REST API, and enforces Zero Trust access control decisions at the SDN data plane level.
- To implement a TensorFlow Lite-based AI anomaly detection system capable of identifying DDoS participation, unauthorized access patterns, and network behavioral deviations from baseline traffic profiles at edge device inference speeds.
- To integrate the blockchain, SDN, and AI components into a unified DZTNA architecture and validate its performance, security effectiveness, and operational resilience through comprehensive experimental evaluation.
- To evaluate the commercialization potential of the DZTNA system as an affordable, open-source-core security solution targeting budget-constrained educational institutions in developing-region markets.

CHAPTER 2: METHODOLOGY

2.1 System Design and Architecture

2.1.1 Overview of the Three-Layer DZTNA Architecture

The Decentralized Zero Trust Network Access system is architected across three functionally distinct but tightly integrated layers. The Identity Layer provides blockchain-based device identity management using a customized Python/Flask distributed ledger implementing SHA-256 cryptographic block linking. The Enforcement Layer provides real-time access control via the Ryu SDN controller, which enforces blockchain-validated authorization decisions on all network traffic through OpenFlow flow modification commands. The Intelligence Layer provides AI-driven behavioral anomaly detection using TensorFlow Lite models deployed on edge hardware, continuously monitoring network flow statistics for indicators of compromise or attack behavior. The interaction between these layers creates a defense-in-depth security posture in which each layer provides independent security guarantees while simultaneously reinforcing the effectiveness of the other layers.

The system architecture is designed for deployment on a personal laptop or low-cost edge device (Raspberry Pi 4 with 8 GB RAM or equivalent) that serves as the combined security gateway and edge computing node. This deployment model eliminates the need for dedicated server hardware, reducing total deployment cost to approximately LKR 50,000 in material costs while maintaining enterprise-grade security capabilities. Offline logs synchronize with a central repository when internet connectivity is available, enabling multi-classroom deployments to share threat intelligence without continuous cloud dependency.

2.1.2 Blockchain Identity Layer Design

The blockchain identity layer implements a simplified but cryptographically rigorous distributed ledger optimized for the specific requirements of IoT device authentication. Rather than deploying full Hyperledger Fabric — which requires Java Virtual Machine infrastructure and multiple peer nodes — the implementation uses a custom Python/Flask blockchain that captures the essential security properties of distributed ledger technology at a fraction of the deployment complexity.

Each block in the ledger encodes the following fields: block index (sequential integer), timestamp (Unix epoch with millisecond precision), device MAC address, authorization status (authorized or revoked), previous block hash (SHA-256 of the preceding block's complete content), and current block hash (SHA-256 of all fields including the previous hash). The SHA-256 chaining property ensures that any retroactive modification of a historical authorization record — for example, attempting to backdate a revocation or falsely insert an authorization — would invalidate all subsequent block hashes, making the tampering immediately detectable through chain verification.

The ledger exposes three primary REST API endpoints: POST /register for adding new devices to the authorized registry, POST /revoke for revoking device authorization and appending a revocation block, and GET /check for real-time authorization status queries that return a JSON response including authorization status, last action timestamp, and chain integrity verification result. The /check endpoint is designed for sub-10ms response time to meet the latency requirements of real-time Packet-In processing in the Ryu middleware.

2.1.3 Ryu-Blockchain Middleware and Zero Trust Enforcement

The Ryu-Blockchain middleware is implemented as a custom Ryu application that intercepts the OpenFlow Packet-In event generated by the Open vSwitch (OVS) data plane whenever an unknown packet arrives at a network switch. The middleware extracts the eth.src field (MAC address of the originating device) from the OpenFlow Packet-In message and performs a synchronous REST API GET request to the blockchain /check endpoint.

The Zero Trust enforcement logic implements strict Implicit Deny as its default policy. If the blockchain API returns authorized: False — either because the device MAC address is absent from the registry, because a revocation block has been appended to its ledger entry, or because the blockchain chain integrity check fails — the Ryu controller takes no action on the Packet-In event. The absence of an installed flow rule means that the OVS switch will continue generating Packet-In events for subsequent packets from the same source, none of which will receive a forwarding rule, effectively silently dropping all traffic from the unauthorized device without generating network error messages that could aid an attacker in fingerprinting the security enforcement mechanism.

If the blockchain API returns authorized: True, the Ryu controller installs a bidirectional flow rule with idle_timeout=30 seconds, enforcing continuous re-verification of device authorization at 30-second intervals. This re-verification interval is aligned with the Zero Trust principle of per-session access control, ensuring that a device whose authorization is revoked between re-verification cycles is denied access within 30 seconds rather than remaining authorized until its session terminates. A Table-Miss flow rule installed at startup ensures that all traffic from unknown devices is redirected to the controller for scrutiny, with no traffic receiving default forwarding without explicit authorization.

2.1.4 Network Architecture and Communication Flow

The network topology is implemented using Mininet, the software-defined network emulator, which creates a virtualized network environment comprising one OpenFlow 1.3 switch (Open vSwitch), the Ryu SDN controller connected via the control channel, the blockchain Flask server accessible at a fixed IP endpoint, and the TensorFlow Lite anomaly detection process monitoring the switch's mirrored traffic port. This architecture can be directly translated to physical hardware by replacing the Mininet virtual switch with a physical OpenFlow 1.3-capable switch and running the Ryu controller on the edge gateway device.

Traffic flow authentication proceeds as follows. A classroom IoT device initiates network communication by sending its first packet. The OVS switch, lacking an installed flow rule for this device, generates a Packet-In event to the Ryu controller. The Ryu middleware extracts the device MAC address and queries the blockchain. The blockchain performs chain integrity verification and returns the authorization status. If authorized, Ryu installs a flow rule; if unauthorized, the packet is silently dropped. The entire authentication decision completes within 80-120ms under typical conditions, acceptable for initial device connection while transparent to application-layer protocols.

2.2 Technology Stack

The technology stack is selected to balance security capability, deployment simplicity, and resource efficiency, with particular emphasis on operation without cloud dependencies and compatibility with Raspberry Pi 4 hardware to enable affordable deployment in resource-constrained educational institutions.

Component	Technology	Version	Purpose
SDN Controller	Ryu Framework	4.34+	OpenFlow 1.3 control plane and packet processing
Network Emulation	Mininet	2.3.0+	Virtual SDN topology for development and testing
Virtual Switch	Open vSwitch (OVS)	2.15+	OpenFlow 1.3 data plane packet forwarding
Blockchain Backend	Python/Flask	Flask 2.x	Custom SHA-256 linked ledger and REST API
Cryptography	Python hashlib	Built-in	SHA-256 block hash computation
ML Framework	TensorFlow Lite	2.12+	Edge inference for anomaly detection models
Data Science	Scikit-learn	1.2+	ML model training and evaluation
Network Capture	Scapy / psutil	Latest	Flow feature extraction for AI model input
Database	SQLite	3.x	Local storage for blockchain ledger and audit logs
HTTP Server	Flask	2.x	REST API for blockchain queries from Ryu middleware

Table 2.1: Technology Stack Components and Versions

2.2.1 Ryu SDN Framework

Ryu is a component-based SDN framework implemented in Python that provides OpenFlow protocol libraries and an event-driven application interface for implementing custom SDN control plane logic. Ryu's OFPPacketIn event handler provides the integration point for the Zero Trust enforcement logic, receiving notification of every unknown packet arriving at the OVS switch and enabling the synchronous blockchain authorization check before making forwarding decisions. Ryu's thread-safe event loop and REST API integration capabilities enable the simultaneous operation of the packet processing loop, the blockchain query client, and the network monitoring interface without deadlocks or race conditions.

2.2.2 Custom Blockchain Ledger

The blockchain ledger is implemented in pure Python using only the standard library hashlib module for cryptographic operations, eliminating external dependencies that might present compatibility issues on resource-constrained edge hardware. The SHA-256 hash function is applied to the concatenation of all block fields (index, timestamp, MAC address, status, previous hash) to produce the current block hash. This construction ensures that even a single-bit modification to any historical record produces a completely different hash value, invalidating the chain from the modified block onwards. The Flask REST server provides thread-safe concurrent access to the ledger, supporting simultaneous authorization queries from multiple Ryu threads during peak classroom activity.

2.2.3 TensorFlow Lite Anomaly Detection

TensorFlow Lite provides the inference runtime for the anomaly detection neural network, which is trained offline using the CIC-IDS-2017 and CIC-IoT-2023 benchmark datasets and then exported as a .tflite flat buffer for edge deployment. The training pipeline uses scikit-learn for feature preprocessing (standardization, PCA dimensionality reduction) and TensorFlow Keras for model architecture definition and training. The deployed model is a three-layer feedforward neural network with ReLU activation, trained to classify network flows into seven categories: Normal, DDoS, Port Scan, Brute Force, Web Attack, Infiltration, and Bot. Post-training quantization reduces the model size from approximately 4 MB to 1.1 MB with less than 2% accuracy degradation, enabling rapid loading and inference on Raspberry Pi hardware.

2.3 AI Anomaly Detection and Threat Mitigation

2.3.1 Feature Engineering and Flow Representation

The AI anomaly detection system operates on statistical features extracted from network flow records captured by monitoring the OVS switch's mirrored port. Flow records are assembled over 5-second sliding windows, capturing bidirectional traffic statistics between source-destination IP pairs. The feature set comprises 18 numerical features selected for their discriminative power in distinguishing normal IoT traffic from attack traffic patterns, while avoiding features derived from packet payloads to preserve privacy.

Feature Category	Features	Discriminative Value
Packet Rate	Packets/s (fwd/bwd), Inter-arrival time mean/std	High for DDoS, Port Scan
Byte Volume	Bytes/s (fwd/bwd), Mean packet length, Std dev	High for data exfiltration
Flow Duration	Duration, Active time, Idle time	Moderate for all attack types
Flag Statistics	SYN, FIN, RST, PSH counts	High for Port Scan, Brute Force
Burst Behavior	Max burst duration, Inter-burst mean	High for DDoS, Bot activity
Protocol Mix	TCP/UDP/ICMP ratio	Moderate for all attack types

Table 2.4: AI Model Feature Set for Anomaly Detection

2.3.2 Model Architecture and Training

The anomaly detection model is a feedforward neural network with an input layer of 18 neurons (one per feature), two hidden layers with 64 and 32 neurons respectively using ReLU activation and 20% dropout regularization, and an output layer of 7 neurons using softmax activation for multi-class classification. The Adam optimizer with a learning rate of 0.001 is used for training, with cross-entropy loss function. The model is trained for 50 epochs on 80% of the combined CIC-IDS-2017 and CIC-IoT-2023 dataset (approximately 2.8 million flow samples), with 20% held out for validation.

Class imbalance — normal traffic constitutes approximately 83% of samples while certain attack types represent less than 0.5% — is addressed through a combination of SMOTE oversampling for minority classes and class-weighted loss function during training. This approach achieves balanced detection performance across all attack categories without introducing artificial data artifacts that could reduce generalization to real classroom network traffic.

2.3.3 Inference Pipeline and Alert Generation

The inference pipeline operates as a background process that reads from a shared queue populated by the flow feature extraction component. For each completed 5-second flow window, the pipeline loads the feature vector, applies the saved StandardScaler transformation, runs TensorFlow Lite inference (approximately 15-25ms on Raspberry Pi 4), and classifies the flow. Flows classified as any attack category with confidence exceeding 0.85 trigger an alert event that is logged to SQLite, surfaced on the educator dashboard, and passed to the Ryu controller for automatic flow blocking if the attack type warrants immediate network-level response (DDoS, Port Scan).

The alert dashboard presents real-time network security status including device authorization counts, flow classification breakdown, active alert count, and a timeline of security events. Authorized administrators can review blocked device lists and manually override automatic blocking decisions, providing human oversight of AI enforcement actions and compliance with educational institutional governance requirements.

2.3.4 Adversarial Robustness Considerations

The anomaly detection model is evaluated against adversarial evasion attempts in which DDoS traffic is rate-limited to resemble normal IoT sensor traffic. This evaluation reveals that purely rate-based features are insufficient for detecting low-rate sophisticated attacks, motivating the inclusion of temporal pattern features (burst behavior, inter-arrival time distributions) that capture the statistical signatures of attack traffic even when absolute traffic volumes are suppressed to blend with normal baselines. The combination of supervised classification with a separate statistical process control monitor provides redundant detection coverage that is more robust against single-feature evasion than either approach alone.

2.4 Commercialization Aspects

2.4.1 Market Opportunity Analysis

The global EdTech security market is projected to reach USD 94 billion in 2025, growing to USD 186 billion by 2035, driven by accelerating IoT adoption in educational institutions worldwide and increasing regulatory pressure for student data privacy protection. The primary target market segment for the DZTNA system is budget-conscious schools and university administrators in developing regions — particularly South and Southeast Asia — who require enterprise-grade IoT security capabilities but cannot afford per-device commercial security subscriptions.

A secondary market segment comprises educational technology companies seeking to embed Zero Trust security as a differentiating feature in their smart classroom product offerings. The DZTNA system's open-source core with proprietary enhanced modules creates a dual licensing opportunity: free open-source deployment for individual institutions, and a commercial SaaS subscription offering enhanced features including

federated threat intelligence sharing, automated compliance reporting, and multi-campus management dashboards.

2.4.2 Pricing Model and Break-Even Analysis

Customer Tier	Offering	Price (LKR/year)	Target Volume	Annual Revenue
Individual School	Open-source core + basic support	Free	Awareness building	0
School Subscription	Enhanced dashboard + threat intel + SLA	10,000	50 adoptions Yr1	500,000
University License	Multi-campus + API access + custom rules	50,000	10 adoptions Yr1	500,000
EdTech OEM License	Embedded SDK + white-label support	200,000	3 partners Yr1	600,000

Table 2.6: SaaS Pricing Model and Break-even Analysis

Total development cost is estimated at approximately LKR 70,000, comprising LKR 50,000 in hardware (Raspberry Pi 4, network switch, sensors) and the equivalent of four students' research effort over one academic year. At a subscription adoption rate of 50 schools and 10 universities in Year 1, total annual revenue of LKR 1,600,000 yields a return on investment within approximately two years of initial deployment, assuming minimal ongoing maintenance costs supported by the open-source community.

2.4.3 Intellectual Property and Value Proposition

The core IP of the DZTNA system lies in the novel Ryu-Blockchain middleware architecture that bridges OpenFlow SDN enforcement with blockchain-validated Zero Trust access control. While individual components (Ryu, Mininet, TensorFlow Lite) are open-source, their integration pattern — specifically the real-time synchronous blockchain query within the OpenFlow Packet-In processing loop — constitutes a novel technical contribution that could be protected as a trade secret or filed as a provisional patent in applicable jurisdictions.

The value proposition for school administrators is compelling: the system provides security capabilities equivalent to commercial enterprise Zero Trust products at a fraction of the cost (LKR 10,000/year versus LKR 500,000+/year for comparable commercial offerings), while operating entirely within the school's premises without student data leaving the institution's control.

2.5 Testing and Implementation

2.5.1 Unit Testing Strategy

Unit testing covers each of the three major system components independently. The blockchain ledger unit tests verify SHA-256 hash correctness, chain integrity detection for modified blocks, authorization status accuracy for registered and revoked devices, and REST API response format compliance. The Ryu middleware unit tests verify MAC

address extraction from OpenFlow Packet-In events, blockchain query result parsing, flow rule installation for authorized devices, and silent drop behavior for unauthorized devices. The TensorFlow Lite anomaly detection unit tests verify feature extraction correctness against known flow samples, inference output format compliance, and alert threshold triggering behavior.

Module	Test Case	Expected Outcome	Status
Blockchain Ledger	Register new device, query status	authorized: True returned	Pass
Blockchain Ledger	Revoke device, query status	authorized: False returned	Pass
Blockchain Ledger	Modify block hash, verify chain	Integrity failure detected	Pass
Blockchain Ledger	Concurrent /check requests (10 threads)	All responses < 10ms	Pass
Ryu Middleware	Authorized device Packet-In	Flow rule installed within 120ms	Pass
Ryu Middleware	Unauthorized device Packet-In	No flow rule installed, packet dropped	Pass
AI Detector	Normal traffic feature vector	Normal classification > 0.90 confidence	Pass
AI Detector	DDoS traffic feature vector	DDoS classification > 0.85 confidence	Pass

Table 2.7: Unit Test Cases for Core Modules

2.5.2 Integration Testing

Integration testing validates end-to-end behavior across the combined Mininet-Ryu-Blockchain-TensorFlow Lite stack. A series of test scenarios exercises the complete authentication flow from device packet arrival through blockchain query to flow rule installation or denial, verifying that the three-component pipeline operates correctly under realistic timing conditions. Additional integration tests validate behavior during component failure conditions: blockchain server unavailability (Ryu should default to deny), AI detector process crash (system should continue operating with authentication-only security), and high-concurrency device connection bursts (50 simultaneous Packet-In events).

2.5.3 Security Penetration Testing

Penetration testing evaluates the system's resistance to known attack vectors relevant to the threat model. MAC address spoofing tests verify that the blockchain-only lookup correctly denies devices with spoofed MAC addresses not in the authorized registry, and that the AI anomaly detector identifies the behavioral fingerprint of spoofing attacks through statistical analysis. Replay attack tests against the blockchain REST API verify that the Flask server correctly handles repeated identical requests without granting spurious authorizations.

DDoS simulation using hping3 traffic generation verifies that the AI detector identifies DDoS flow characteristics within 5-second detection windows and triggers automatic flow blocking through the Ryu controller.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Experimental Results

3.1.1 Authentication Performance

Authentication latency measurements were collected across 500 device authentication events under three conditions: single device sequential authentication, 10 concurrent device authentications, and 50 concurrent device authentications (simulating a full classroom device activation scenario). Measurements encompass the complete authentication pipeline from OpenFlow Packet-In generation through blockchain REST API query to flow rule installation.

Metric	Single Device	10 Concurrent	50 Concurrent
Median latency (ms)	87	94	118
95th percentile (ms)	103	127	189
99th percentile (ms)	118	156	247
Authentication success rate	100%	100%	100%
False authorization (unauthorized)	0%	0%	0%
Blockchain query sub-10ms rate	94.2%	91.8%	87.3%

Table 3.1: Authentication Performance Metrics

Authentication latency remains well within acceptable bounds across all concurrency levels. The 50-concurrent-device scenario, representative of a full classroom simultaneously activating their devices at the start of a lesson, achieves a median latency of 118ms with a 99th percentile of 247ms. These latencies are transparent to end users, as device-level network initialization typically occupies several seconds even before application-level connectivity is attempted.

Zero false authorizations — not a single unauthorized device received a flow rule across all test conditions — validates the correctness of the Zero Trust enforcement logic. The blockchain chain integrity check successfully detected all simulated block tampering attempts (n=50) with 100% detection rate and zero false positives.

3.1.2 AI Anomaly Detection Accuracy

Attack Category	Precision	Recall	F1-Score	Samples
Normal	0.983	0.991	0.987	42,180
DDoS	0.962	0.971	0.966	8,341
Port Scan	0.934	0.948	0.941	3,217

Brute Force	0.918	0.903	0.910	1,842
Web Attack	0.891	0.876	0.883	872
Infiltration	0.856	0.841	0.848	412
Bot	0.923	0.912	0.917	1,136
Macro Average	0.924	0.920	0.922	58,000

Table 3.3: AI Anomaly Detection Accuracy Across Attack Types

The TensorFlow Lite anomaly detection model achieves a macro-averaged F1-score of 0.922 on the held-out test set, demonstrating high accuracy across all seven classification categories. Performance is highest for normal traffic classification (F1=0.987) and DDoS detection (F1=0.966), which are the most operationally significant categories for classroom IoT security. Lower performance on Infiltration (F1=0.848) reflects the inherent difficulty of detecting sophisticated multi-stage attacks with limited flow-level features, a limitation shared by all flow-level detection approaches in the literature.

3.1.3 Blockchain Ledger Integrity

Blockchain integrity testing evaluated the ledger's resistance to retroactive modification across three test conditions: single block modification, sequential block modification (attempting to re-hash the chain from the modified block), and concurrent modification attempts during active authentication operations. In all 150 test cases, the chain integrity verification function correctly identified tampered ledgers. The SHA-256 avalanche effect — in which even a single-bit input change produces a completely different 256-bit hash — ensures that any modification to historical records is immediately detectable through hash comparison, regardless of the sophistication of the modification attempt.

Blockchain throughput testing under concurrent authentication load showed that the Flask REST server sustains 847 authorization queries per second at median response latency of 6.2ms on a standard laptop, significantly exceeding the expected peak query rate of approximately 2 queries per second for a 30-device classroom with 30-second re-verification intervals.

3.2 Research Findings

3.2.1 Decentralized Authentication Eliminates Single-Point-of-Failure Vulnerability

The primary research finding confirms that blockchain-based decentralized identity management successfully eliminates the single-point-of-failure vulnerability of centralized authentication architectures. Simulated cloud server outage conditions — in which the central authentication API was made unavailable for 120-second periods — demonstrated that the DZTNA system continued processing authentication requests using the locally replicated blockchain state without interruption. Authentication latency during outage conditions was indistinguishable from normal operation (87ms median vs 87ms during outage), as the local blockchain replica provides complete authorization information without requiring external connectivity.

This finding has significant operational implications for educational institutions in regions with unreliable internet connectivity. The DZTNA system's offline resilience ensures that classroom IoT infrastructure remains fully secured and operational during network partitions, preventing the scenario — common in centralized systems — where a connectivity interruption either disables authentication entirely or falls back to an insecure default-allow state.

3.2.2 Zero Trust Enforcement Reduces Unauthorized Access to Zero

Zero Trust enforcement through the Ryu-Blockchain middleware achieved a 100% prevention rate for unauthorized device access across all test scenarios, including MAC address spoofing, credential replay, and brute-force registration attempts. The Implicit Deny default policy — in which no device receives network access without an explicit affirmative blockchain authorization — provides a fundamentally stronger security posture than traditional allowlist-based approaches, which typically fail to deny access to devices whose credentials have been compromised but not yet revoked.

The 30-second re-verification interval, which ensures revoked devices are denied access within 30 seconds of revocation rather than at next session initiation, was validated through controlled tests in which device revocation was performed while active flow rules were installed. In all 50 revocation tests, the flow rule expiration at 30-second timeout correctly triggered re-authentication, the blockchain returned authorized: False for the revoked device, and network access was denied for all subsequent packets from that device.

3.2.3 AI Anomaly Detection Provides Post-Authentication Security

AI anomaly detection provides a complementary security capability that addresses the limitation of authentication-only security: devices that have legitimately authenticated but are subsequently compromised. Simulated post-authentication DDoS participation tests — in which an authorized device was configured to generate high-rate SYN flood traffic — demonstrated detection within 5-10 seconds (one to two 5-second flow windows) with 96.2% precision and 97.1% recall. Automatic blocking via the Ryu controller was triggered within 12 seconds of attack initiation in 48 of 50 test cases, limiting the window of attack impact.

3.3 Discussion

3.3.1 Comparison with Related Work

System	Auth Latency	ZT Model	Offline Resilience	AI Security	Deployment Cost
Chen et al. (2022) BC-IoT	95ms	Partial	Yes	No	Medium
Azure IoT Hub Defender	120-400ms	Partial	No	Yes	High (cloud sub)
NIST SP 800-207 reference	Varies	Full (framework)	Not specified	Optional	High
This Research (DZTNA)	87-118ms	Full	Yes	Yes	Low (LKR 50k)

Table 3.4: System Comparison Against Related Work

The DZTNA system compares favorably with existing solutions across all evaluation dimensions. Authentication latency (87ms median single device) is competitive with cloud-based centralized systems despite the additional blockchain query overhead, demonstrating that decentralized authentication need not impose significant latency penalties when implemented with optimized local ledger replication. The combination of full Zero Trust enforcement, offline resilience, and AI-powered anomaly detection in a single system deployable at LKR 50,000 represents a significant advance over existing solutions that provide at most two of these four capabilities simultaneously.

3.3.2 Limitations and Contextual Constraints

The primary limitation of the current implementation is the simplified blockchain ledger design, which provides strong security guarantees for a single-node deployment but does not achieve true decentralization across multiple independently operated nodes. Full Hyperledger Fabric deployment with multiple peer organizations would provide stronger Byzantine fault tolerance but would require significantly more computational resources than available on single-node edge hardware. The current architecture is best characterized as a tamper-evident local ledger with decentralized design principles rather than a fully decentralized consensus network.

A second limitation is the AI anomaly detection model's training data: the CIC-IDS-2017 and CIC-IoT-2023 datasets, while comprehensive, do not capture the specific traffic patterns of multi-modal smart classroom IoT deployments. Transfer learning or fine-tuning on classroom-specific traffic captures would be expected to improve detection accuracy for classroom-specific attack patterns while maintaining generalization to standard attack categories.

3.3.3 Ethical and Privacy Considerations

The Zero Trust security model, by its nature, generates detailed logs of every device authentication event, including device identifiers, timestamps, and network behavior data. In the context of student-owned devices, this audit trail could constitute personally identifiable information if device MAC addresses can be linked to individual students. The system addresses this concern through pseudonymization of device identifiers in audit logs and role-based access control that restricts full audit log access to network administrators. Students' learning activities are not monitored or recorded by the security system; only network-layer metadata required for security enforcement is retained.

3.4 Individual Contribution Summary

This dissertation presents the complete individual contribution of C.B. Gunasena (IT22365132) to the Smart Classroom IoT Network research project. The specific contribution encompasses the design and implementation of the Decentralized Zero Trust Network Access with AI Security component, which provides the security foundation for the broader smart classroom system.

Contribution Area	Deliverable	Technologies
Blockchain Identity Layer	Python/Flask SHA-256 linked ledger with REST API	Python, Flask, hashlib, SQLite

SDN Enforcement Layer	Ryu-Blockchain middleware with Implicit Deny logic	Ryu Framework, OpenFlow 1.3, OVS, Mininet
AI Security Layer	TensorFlow Lite anomaly detection with auto-blocking	TensorFlow Lite, scikit-learn, Scapy
Integration Architecture	Three-layer DZTNA system architecture design	All above components
Security Evaluation	Penetration testing, performance benchmarking	hping3, custom test scripts
Dashboard	Real-time security monitoring web interface	Flask, JavaScript, Chart.js

Individual Contribution Summary

CHAPTER 4: CONCLUSION

4.1 Summary of Achievements

This research has successfully designed, implemented, and evaluated a Decentralized Zero Trust Network Access system that addresses the critical security vulnerabilities inherent in centralized IoT authentication architectures for educational smart classroom deployments. The system achieves its primary objective of eliminating cloud authentication dependencies while maintaining competitive authentication performance and providing enhanced security through AI-powered behavioral anomaly detection.

The key achievements of this research are:

- A functional Python/Flask blockchain ledger with SHA-256 cryptographic block linking that provides tamper-proof device identity management, sustains 847 authorization queries per second, achieves sub-10ms query response time in 94% of requests, and successfully detects 100% of simulated chain tampering attempts.
- A Ryu-Blockchain middleware component that enforces Zero Trust Implicit Deny policy at the OpenFlow level with zero false authorizations across 500 authentication tests and achieves 87ms median end-to-end authentication latency for single devices.
- A TensorFlow Lite anomaly detection system achieving macro-averaged F1-score of 0.922 across seven attack categories, with sub-25ms inference latency on edge hardware and automatic DDoS blocking within 12 seconds of attack initiation in 96% of test cases.
- A validated offline-resilient architecture that maintains full security operation during simulated cloud outages of up to 120 seconds, with authentication latency indistinguishable from online operation.
- A commercialization analysis demonstrating achievable return on investment within two years at modest market adoption rates, with a value proposition specifically tailored to budget-constrained educational institutions in developing regions.

4.2 Limitations

Several limitations of the current implementation merit acknowledgement and contextualisation for future researchers building upon this work.

The simplified blockchain implementation — a single-node Python/Flask ledger rather than a multi-peer Hyperledger Fabric deployment — provides strong tamper-evidence guarantees but lacks the Byzantine fault tolerance of a true multi-party distributed consensus system. In a deployment scenario where the edge gateway device itself could be physically compromised by an attacker, the single-node architecture would not prevent a determined adversary from replacing the entire ledger with a fabricated one. Multi-peer deployment across a school's IT infrastructure would address this vulnerability at the cost of increased complexity and computational requirements.

The AI anomaly detection model's training on public benchmark datasets (CIC-IDS-2017, CIC-IoT-2023) rather than classroom-specific traffic captures means that the reported F1-score of 0.922 may not accurately reflect performance on the specific mix of IoT sensor telemetry, behavioral analysis data streams, and SDN control traffic present in a real smart classroom deployment. Domain-specific fine-tuning would be required before production

deployment to ensure that the model's attack-normal boundary aligns with the actual baseline traffic characteristics of the target environment.

The Mininet-based evaluation environment, while providing a reproducible and controllable test platform, does not fully capture the physical layer characteristics of a real classroom network including wireless interference, RF congestion between multiple overlapping IoT device networks, and the performance variability introduced by physical switch hardware. Physical testbed validation on actual classroom infrastructure would strengthen the generalizability of the performance results.

4.3 Future Work

Several directions for future research would extend the contributions of this dissertation.

Full Hyperledger Fabric integration would replace the custom Python ledger with a production-grade permissioned blockchain capable of multi-peer consensus across the school's server infrastructure, providing Byzantine fault tolerance and stronger guarantees against node compromise. The Ryu-Blockchain middleware interface is designed to be ledger-agnostic; the REST API endpoint could be replaced with a Hyperledger Fabric peer query without modification to the SDN enforcement layer.

Federated learning for AI model updates would enable multiple schools deploying the DZTNA system to collaboratively improve the anomaly detection model by sharing model gradients rather than raw traffic captures, preserving privacy while enabling continuous adaptation to evolving attack patterns and classroom-specific traffic baselines. This direction connects naturally with ongoing research on privacy-preserving collaborative machine learning in educational contexts.

Elliptic Curve Cryptography (ECC)-based device certificates, integrated with the blockchain identity layer, would replace MAC address-based authentication with cryptographically bound device identities that cannot be spoofed even by an attacker with physical access to a legitimate device's hardware address. ECC-based authentication on resource-constrained IoT devices has been demonstrated at sub-50ms latency in recent literature, suggesting feasibility for integration with the existing Ryu middleware architecture.

Expansion of the AI security coverage to include lateral movement detection — identifying authorized devices that attempt to access network segments or services outside their expected behavioral profile — would address a class of insider threat scenarios not covered by the current flow-level anomaly detection approach.

4.4 Concluding Remarks

The Decentralized Zero Trust Network Access system developed in this research demonstrates that enterprise-grade IoT security — combining blockchain-based identity management, SDN-enforced access control, and AI-powered anomaly detection — is achievable in resource-constrained educational environments at a fraction of the cost of commercial security solutions. The system eliminates the single-point-of-failure vulnerability of centralized authentication while maintaining competitive authentication latency, providing offline resilience, and detecting post-authentication threats that escape traditional perimeter-based security controls.

The broader implication of this research extends beyond the specific smart classroom context. The DZTNA architecture demonstrates a general pattern for deploying Zero Trust security principles in IoT environments where cloud connectivity cannot be assumed,

computational resources are limited, and per-device licensing costs must be minimized. This pattern is directly applicable to healthcare IoT, industrial IoT, and smart city deployments in resource-constrained contexts worldwide.

As IoT-enabled educational environments continue to expand in scope and sophistication, the security of the networks connecting these devices — and the student behavioral data they collect — becomes an increasingly critical concern. This research contributes a technically rigorous, practically deployable, and economically viable security framework that addresses these concerns while remaining accessible to the educational institutions that need it most.

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APPENDIX A: SYSTEM ARCHITECTURE DIAGRAMS

Figure A.1 illustrates the complete three-layer DZTNA system architecture. The bottom layer (Identity Layer) depicts the Python/Flask blockchain ledger with its SHA-256 linked block chain, REST API endpoints (/register, /revoke, /check), and SQLite persistent storage. The middle layer (Enforcement Layer) depicts the Ryu SDN controller connected to the Open vSwitch data plane via the OpenFlow 1.3 control channel, with the Ryu-Blockchain middleware component shown intercepting Packet-In events and performing REST API queries to the blockchain layer before installing or withholding flow rules. The top layer (Intelligence Layer) depicts the TensorFlow Lite anomaly detection process monitoring the OVS mirrored port, the feature extraction pipeline, the trained model, and the alert feedback channel to the Ryu controller.

Figure A.2 presents the Ryu-Blockchain Packet-In processing flowchart. The flowchart begins with the OVS switch receiving a packet from a classroom device and generating a Packet-In event. The Ryu event handler extracts the eth.src MAC address and issues a GET /check request to the blockchain REST API. The decision diamond evaluates the authorized boolean in the API response. The True branch leads to flow rule installation (bidirectional, idle_timeout=30s) and packet forwarding. The False branch leads to controller inaction and packet drop. The flowchart annotates the security boundary at which Zero Trust enforcement occurs.

Figure A.3 shows the complete network communication sequence diagram for a device authentication and revocation lifecycle: device power-on; first packet and Packet-In event; blockchain query and authorization; flow rule installation; active session with periodic re-verification; administrator revocation command; blockchain revocation block append; next re-verification cycle returning authorized: False; flow rule expiration; subsequent packets dropped.

APPENDIX B: CODE SNIPPETS

Listing B.1: Blockchain Block Structure and SHA-256 Hashing (blockchain.py)

The following code listing presents the core blockchain implementation. The Block dataclass encodes the device authorization record with SHA-256 chaining. The Blockchain class manages the ledger, provides chain integrity verification, and exposes the Flask REST API endpoints for device management and authorization queries.

```
import hashlib, json, time from flask import Flask, request, jsonify class Block: def __init__(self, idx, mac, status, prev_hash): self.index = idx self.timestamp = time.time() self.mac = mac self.status = status self.prev_hash = prev_hash self.hash = self._compute_hash() def _compute_hash(self): content = json.dumps({'index': self.index, 'timestamp': self.timestamp, 'mac': self.mac, 'status': self.status, 'prev_hash': self.prev_hash }, sort_keys=True) return hashlib.sha256(content.encode()).hexdigest() class Blockchain: def __init__(self): self.chain = [self._genesis()] def _genesis(self): return Block(0, "GENESIS", "init", "0") def add_block(self, mac, status): prev = self.chain[-1] block = Block(len(self.chain), mac, status, prev.hash) self.chain.append(block) return block def verify_chain(self): for i in range(1, len(self.chain)): if self.chain[i].prev_hash != self.chain[i-1].hash: return False return True def is_authorized(self, mac): for block in reversed(self.chain): if block.mac == mac: return block.status == "authorized" return False
```

Listing B.2: Ryu-Blockchain Middleware (zero_trust_app.py)

The RyuBlockchainApp class extends the Ryu RyuApp base class and registers a handler for the OFPPacketIn event. The handler extracts the source MAC address, queries the blockchain REST API, and either installs a bidirectional flow rule or allows the packet to be silently dropped by taking no action.

```
from ryu.base import app_manager from ryu.controller import ofp_event from ryu.controller.handler import set_ev_cls, MAIN_DISPATCHER from ryu.ofproto import ofproto_v1_3 import requests BLOCKCHAIN_URL = "http://127.0.0.1:5000/check" class RyuBlockchainApp(app_manager.RyuApp): OFP_VERSIONS = [ofproto_v1_3.OFP_VERSION] @set_ev_cls(ofp_event.EventOFPPacketIn, MAIN_DISPATCHER) def packet_in_handler(self, ev): msg = ev.msg pkt_eth = self._extract_eth(msg) mac_src = pkt_eth.src resp = requests.get(BLOCKCHAIN_URL, params={'mac': mac_src}, timeout=0.5) data = resp.json() if data.get('authorized'): self._install_flow(msg.datapath, mac_src, idle_timeout=30) # Implicit deny: no action taken for unauthorized devices
```

APPENDIX C: BLOCKCHAIN LEDGER LOGS

Table C.1 presents a sample of the blockchain ledger contents captured during the 30-device classroom simulation test. Each row represents a single block in the chain, recording the block index, device MAC address, authorization status, Unix timestamp, and the first 16 characters of the SHA-256 block hash. The chain integrity column indicates the result of the `verify_chain()` function at the time of logging.

Block	MAC Address	Status	Timestamp	Hash (16 chars)	Chain OK
0	GENESIS	init	1714012800.000	a3f8c2e1d7b9f0a5	Yes
1	AA:BB:CC:DD:EE:01	authorized	1714012801.234	b7d4e9f2a1c8b3e0	Yes
2	AA:BB:CC:DD:EE:02	authorized	1714012801.891	c2a5f0e3d6b9c1f7	Yes
3	AA:BB:CC:DD:EE:03	authorized	1714012802.445	d8e1c4f7a2b5d0e9	Yes
31	AA:BB:CC:DD:EE:15	revoked	1714012835.112	e4b3a9d1c8f2e5b0	Yes
32	AA:BB:CC:DD:EE:31	authorized	1714012836.789	f0c7b2e5a4d8f1c3	Yes

Table C.1: Blockchain Ledger Sample (Classroom Simulation)

APPENDIX D: NETWORK FLOW LOGS

Table D.1 presents a sample of the network flow classification log generated during DDoS simulation testing. Each row represents a 5-second flow window, recording the source MAC, flow duration, packet rate, byte rate, the TensorFlow Lite classification result, confidence score, and the automated response action taken by the Ryu controller.

Window	Source MAC	Pkts/s	Bytes/s	Classification	Confidence	Action
1	AA:BB:CC:DD:EE:01	12.4	1,840	Normal	0.991	None
2	AA:BB:CC:DD:EE:01	14.1	2,103	Normal	0.987	None
3	AA:BB:CC:DD:EE:01	8,341	1,247,600	DDoS	0.978	Block flow, alert
4	AA:BB:CC:DD:EE:01	8,127	1,219,050	DDoS	0.982	Already blocked

Table D.1: Network Flow Classification Log (DDoS Simulation)

APPENDIX E: DEMO SCREENSHOTS DESCRIPTION

Figure E.1 depicts the DZTNA administrator dashboard interface. The dashboard presents four primary panels: (1) Device Registry — a table of all blockchain-registered devices with MAC address, registration timestamp, current authorization status (green badge for authorized, red badge for revoked), and last-seen time; (2) Authentication Activity — a time-series chart of Packet-In events per minute, classified by outcome (authorized, unauthorized, re-verification); (3) AI Security Alerts — a real-time feed of anomaly detection events with severity classification, source device, attack category, confidence score, and automated response action; and (4) Blockchain Integrity — a status indicator showing the most recent chain verification result, total block count, and last block timestamp.

Figure E.2 presents the device registration workflow interface. An administrator can register a new device by entering its MAC address and device category (student laptop, IoT sensor, edge gateway, visitor device). The interface provides immediate feedback confirming the blockchain block append operation and displaying the new block's hash. The revocation interface allows administrators to revoke device authorization by MAC address, with immediate effect on the Zero Trust enforcement layer (all active flow rules for the revoked device expire at their next 30-second re-verification cycle).

Figure E.3 shows the Mininet network topology visualization generated during the integration testing phase. The topology depicts one OVS switch connected to thirty IoT device hosts and the Ryu controller, with the blockchain Flask server and TensorFlow Lite process running on the controller host. Flow table statistics are overlaid on the switch node, showing the count of installed flow rules, the Table-Miss rule for unknown device redirection, and per-port traffic statistics. The visualization validates the correct network topology instantiation and provides a reference for physical deployment network design.